

Data Cleaning & Reporting Automation

Project Report

Student Details

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1. Introduction

Data cleaning is an essential step in data analysis. Real-world datasets often contain missing values, duplicate records, and inconsistent information. This project focuses on automating the data cleaning process and generating reports using Python.

The objective of this project is to improve data quality and produce meaningful summaries and visualizations for better decision-making.

2. Objective

- Automate data cleaning using Python.
 - Detect and handle missing values.
 - Remove duplicate records.
 - Generate statistical reports automatically.
 - Create visual summaries using charts.
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3. Tools and Technologies Used

- Python
 - Google Colab
 - Pandas Library
 - Matplotlib Library
 - CSV Dataset
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4. Dataset Description

A customer dataset containing customer information such as age, income, spending habits, and other attributes was used for this project.

The dataset was imported into Python using the Pandas library for preprocessing and analysis.

5. Methodology

Step 1: Data Loading

The dataset was loaded into Python using Pandas.

Step 2: Missing Value Detection

The dataset was examined to identify missing values in different columns.

Step 3: Data Cleaning

Missing values were handled by replacing them with appropriate statistical values. Duplicate records were identified and removed.

Step 4: Data Validation

The cleaned dataset was verified to ensure data consistency and accuracy.

Step 5: Report Generation

Statistical summaries such as count, mean, minimum value, maximum value, and standard deviation were generated automatically.

Step 6: Visualization

A histogram chart was created to visualize the distribution of customer salaries and other numerical features.

6. Results

The following tasks were successfully completed:

- Missing values identified and handled.
- Duplicate records removed.
- Clean dataset generated.
- Statistical summary report created.
- Automated visualization generated.

The cleaned dataset is more reliable and suitable for further analysis and machine learning applications.

7. Conclusion

This project demonstrated the process of automating data cleaning and reporting using Python. Automated preprocessing significantly reduces manual effort and improves data quality. The generated reports and visualizations help in understanding the dataset effectively and support data-driven decision-making.

8. Output Files Generated

1. cleaned_data.csv
 2. summary_report.csv
 3. automation.ipynb
 4. Visualization Charts
 5. Project Report PDF
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9. Screenshots

Screenshot 1:

Dataset Upload

Screenshot 2:

Missing Values Detection

Screenshot 3:

Duplicate Records Removal

Screenshot 4:

Summary Report Generation

Screenshot 5:

Visualization Chart

Screenshot 6:

Generated Output Files

Outcome

Successfully automated data cleaning and reporting workflow using Python, improving data quality and generating meaningful analytical reports.